

## Change Summary

- Project: codexa
- Worktree: local session path omitted
- Branch: codex/backend/codexa-20260712-codexa-agent-ab-eval
- Base: main
- Primary commit: ab7c3c2
- Subject: feat(eval): add external agent A/B harness

## Changed Files

ab7c3c2 feat(eval): add external agent A/B harness

```
.gitattributes | 1 +
README.md | 42 +
benchmarks/agent-ab/README.md | 73 ++
benchmarks/agent-ab/config/codexa-instructions.md | 6 +
benchmarks/agent-ab/config/codexa.mcp.json | 8 +
benchmarks/agent-ab/experiment.json | 45 +
.../environment/Dockerfile | 20 +
.../environment/codexa-mcp-entrypoint.sh | 39 +
.../environment/project/.gitignore | 3 +
.../environment/project/README.md | 10 +
.../environment/project/src/init.py | 1 +
.../environment/project/src/targets.py | 11 +
.../environment/project/tests/testtargets.py | 14 +
.../tasks/path-target-normalization/instruction.md | 13 +
.../solution/solution.patch | 38 +
.../path-target-normalization/solution/solve.sh | 4 +
.../tasks/path-target-normalization/task.toml | 32 +
.../path-target-normalization/tests/Dockerfile | 16 +
.../tests/baseline/.gitignore | 3 +
.../tests/baseline/README.md | 10 +
.../tests/baseline/src/init.py | 1 +
.../tests/baseline/src/targets.py | 11 +
.../tests/baseline/tests/testtargets.py | 14 +
.../tests/candidaterunner.py | 71 ++
.../tests/publictestrunner.py | 53 +
.../tasks/path-target-normalization/tests/test.sh | 7 +
.../path-target-normalization/tests/verify.py | 377 ++++++
docs/guides/agent-ab.md | 285 ++++++
docs/guides/eval-gate.md | 11 +
.../plans/codexa-agent-ab-evaluation-2026-07-12.md | 200 ++++
reports/benchmarks/v0.10.0-agent-ab-pilot-v3.json | 685 ++++++++
scripts/agent-ab-analysis.mjs | 993 ++++++++
scripts/agent-ab.mjs | 1257 ++++++++
tests/agent-ab-analysis-validity.test.ts | 567 ++++++++
tests/agent-ab-report.test.ts | 467 ++++++++
tests/agent-ab-runner.test.ts | 620 ++++++++
tests/agent-ab.test.ts | 307 ++++++
37 files changed, 6315 insertions(+)
create mode 100644 benchmarks/agent-ab/README.md
create mode 100644 benchmarks/agent-ab/config/codexa-instructions.md
```

create mode 100644 benchmarks/agent-ab/config/codexa.mcp.json  
create mode 100644 benchmarks/agent-ab/experiment.json  
create mode 100644 benchmarks/agent-ab/tasks/path-target-normalization/environment/Dockerfile  
create mode 100644 benchmarks/agent-ab/tasks/path-target-normalization/environment/codexa-mcp-entrypoint.sh  
create mode 100644 benchmarks/agent-ab/tasks/path-target-normalization/environment/project/.gitignore  
create mode 100644 benchmarks/agent-ab/tasks/path-target-normalization/environment/project/README.md  
create mode 100644 benchmarks/agent-ab/tasks/path-target-normalization/environment/project/src/init.py  
create mode 100644 benchmarks/agent-ab/tasks/path-target-normalization/environment/project/src/targets.py  
create mode 100644 benchmarks/agent-ab/tasks/path-target-normalization/environment/project/tests/testtargets.py  
create mode 100644 benchmarks/agent-ab/tasks/path-target-normalization/instruction.md  
create mode 100644 benchmarks/agent-ab/tasks/path-target-normalization/solution/solution.patch  
create mode 100644 benchmarks/agent-ab/tasks/path-target-normalization/solution/solve.sh  
create mode 100644 benchmarks/agent-ab/tasks/path-target-normalization/task.toml  
create mode 100644 benchmarks/agent-ab/tasks/path-target-normalization/tests/Dockerfile  
create mode 100644 benchmarks/agent-ab/tasks/path-target-normalization/tests/baseline/.gitignore  
create mode 100644 benchmarks/agent-ab/tasks/path-target-normalization/tests/baseline/README.md  
create mode 100644 benchmarks/agent-ab/tasks/path-target-normalization/tests/baseline/src/init.py  
create mode 100644 benchmarks/agent-ab/tasks/path-target-normalization/tests/baseline/src/targets.py  
create mode 100644 benchmarks/agent-ab/tasks/path-target-normalization/tests/baseline/tests/testtargets.py  
create mode 100644 benchmarks/agent-ab/tasks/path-target-normalization/tests/candidaterunner.py  
create mode 100644 benchmarks/agent-ab/tasks/path-target-normalization/tests/publictestrunner.py  
create mode 100644 benchmarks/agent-ab/tasks/path-target-normalization/tests/test.sh  
create mode 100644 benchmarks/agent-ab/tasks/path-target-normalization/tests/verify.py  
create mode 100644 docs/guides/agent-ab.md  
create mode 100644 docs/plans/codexa-agent-ab-evaluation-2026-07-12.md  
create mode 100644 reports/benchmarks/v0.10.0-agent-ab-pilot-v3.json  
create mode 100644 scripts/agent-ab-analysis.mjs  
create mode 100644 scripts/agent-ab.mjs  
create mode 100644 tests/agent-ab-analysis-validity.test.ts  
create mode 100644 tests/agent-ab-report.test.ts  
create mode 100644 tests/agent-ab-runner.test.ts  
create mode 100644 tests/agent-ab.test.ts

#### Verification

- git diff --check: passed
- PYTHONDONTWRITEBYTECODE=1 npm run check: passed

- Codexa post-edit-review: completed; local artifact omitted
- Codexa test-plan: completed; local artifact omitted
- git diff --cached --check: passed
- staged safety scan: passed

## Notes

Adds a source-only Harbor 0.18.0 control/treatment harness for measuring Codexa's effect on coding-agent task completion with an external no-network verifier. Archives the non-confirmatory GPT-5.6 Sol v3 pilot: both arms completed 2/2, while treatment used 3.88x reported cost, 2.28x agent time, and 8.15x input tokens. The docs explicitly limit this to one easy task and disclose that the registered control-character oracle covered ASCII C0 plus DEL, not Unicode C1. No Codexa production src, dependency, bin, manifest, lockfile, or packed evaluator surface changes.